

UNIT 6 Health and Exercise (II)

Health and Exercise (II)

Unit Goals

- Understand figures & tables
- Use comparison and contrast
- Use passive voice in academic writing
- Use hedging in academic writing
- Write a results section
- Write a discussion section

- Lesson 1

- Lesson 2



Lesson 1

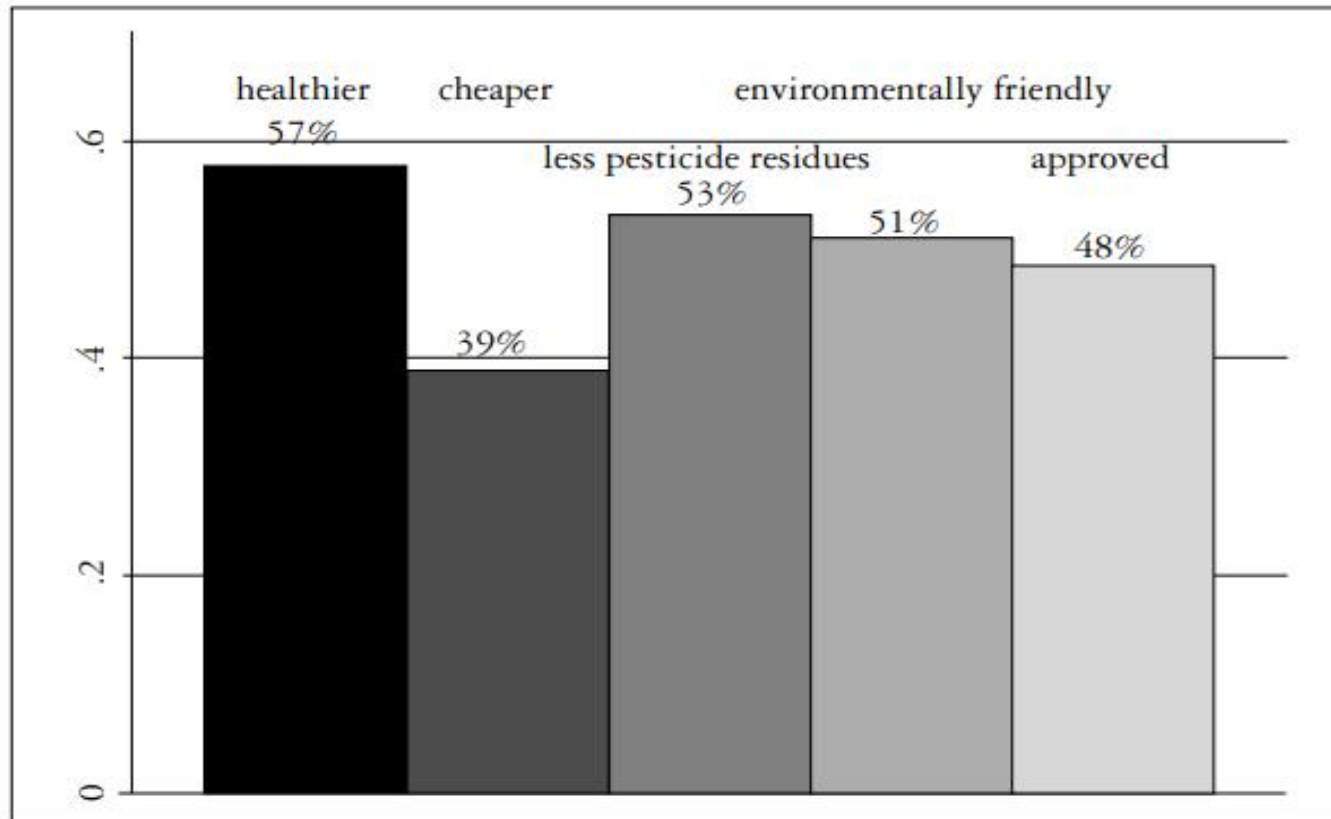
- **Lead-in**
- **Reading**
- **Comprehension Check**
- **Micro-skills**
- **Macro-skills**



Lesson 1 Lead-in

Look at the survey on genetically modified food (Figure 2) in some European countries. Discuss the following questions with a partner.

Figure 2. I would buy genetically modified food if...





Lesson 1 Lead-in

Look at the survey on genetically modified food (Figure 2) in some European countries. Discuss the following questions with a partner.

1. What does the bar chart show?

It shows the percentage of Europeans buying genetically modified food for different reasons.

2. According to the data, what are the major findings of the survey?

57% of Europeans would buy GM food if it were healthier. Only 39% of Europeans would buy GM food if it were cheaper than other foods.



Lesson 1 Lead-in

3. What are the most important elements influencing people's purchasing intention?

The health and environmental related reasons are the most important elements influencing people's purchasing intention.

4. Are you willing to taste or buy GM foods (e.g. a GM apple)?

Open.



Lesson 1

Reading

Text A

Psychological Determinants of Willingness to Taste and Purchase Genetically Modified Food (Extract 2)

3. Results

3.1. Willingness to Purchase and Taste GM Food

Para. 1 Results showed that 48% of the sample stated that they would buy GM food when it became available (purchasers), compared to 49% who would not (nonpurchasers), with three nonresponses. Nonpurchasers demonstrated significantly higher (i.e. negative) ratings for both GM plants and GM animals on most attitude scales. For example, nonpurchasers viewed GM food as significantly more dreaded, more risky, and of a higher risk to future generations. Table 1 illustrates the magnitude of differences between the two groups. There were large differences in ratings of whether genetic modification of plants and animals for food is dreaded and viewed as unethical, with nonpurchasers' ratings at least 22% higher on these scales for both plants and animals. This magnitude of difference in ratings between the two groups was also observed in judgments on the risk of GM (21.3% for plants and 21.1% for animals), and also on the risk of GM to future generations (17.9% for plants and 26.1% for animals).



Lesson 1

Reading

On the disgusting — not disgusting scale, nonpurchasers again rated GM more negatively (20.9% and 17.6%, respectively). No differences were observed between purchasers and nonpurchasers' ratings in regard to the perceived dullness — vividness of the topic of GM food, and both groups felt that GM animals and plants affected them personally.

Para. 2 Strikingly, 93% of our sample willingly tasted the GM apple. Of the 7% that refused, four were concerned about possible consequences to their health. The other three reasons were concerns about the environment, apprehension induced by media coverage, and lack of knowledge about GM food (all $n = 1$). The number of participants refusing to taste the GM apple was so small that it made it impossible to compare these groups on other key variables (e.g. attitude measures) in the study. Furthermore, no differences were observed on any outcome measures between participants exposed to different versions of the study booklet, one of which said the apple had been modified to prevent tooth decay (health benefit), the other that it was to prevent rotting (money-saving benefit).

Para. 3 A mixed-design ANOVA revealed there were no significant differences between purchasers and nonpurchasers in regard to appearance ratings for the “organic”, “traditional” and “GM” apples. In regard to tastiness, despite the fact that all the apples were the same (traditionally grown Red Delicious apples), nonpurchasers' ratings of what they believed to be the GM apple were lower than purchasers' ratings (means 4.02 vs. 4.60, respectively), but this difference did not reach statistical significance ($p = 0.09$).



Lesson 1

Reading

3.2. Affect and Judgments about GM Food

3.2.1. Integral Affect and Willingness to Purchase GM Food

Para. 4 The fact that nonpurchasers' ratings were significantly higher (i.e. more negative) for our measures of feelings about GM food (dread and disgust — see Table 1) is interesting, but do such feelings toward GM plants and animals contribute more than the other attitude variables toward willingness to purchase GM food? A forward (likelihood ratio) stepwise logistic regression was conducted on the attitude and affect measures to investigate this. The 20 attitude and affect variables that differed significantly between purchasers and nonpurchasers (displayed in Table 1) were entered into the analysis. The results of this analysis may be seen in Table 2. Three variables emerged from this analysis as significant predictors of the willingness to purchase GM food. These were (1) dread of GM animals, (2) dread of GM plants, and (3) the belief that GM animals pose a high risk to future generations. This procedure revealed that our model can account for 46% of the variance in willingness to purchase GM food (so approximately half of what determines willingness to purchase GM food is still unknown).

3.2.2. Effects of Negative Incidental Affect (Stress and Worry) and Arousal on Perceived Risks from GM Food



Lesson 1 Reading

Para. 5 Finally, we examined whether the state measures of stress and arousal, and the trait worry measures taken, affected risk-related judgments about GM food. For each measure a median split of scores was performed in order to divide participants into two groups based on their responses. Hence, for each measure (stress, arousal, and worry), there were two groups (high vs. low). We calculated participants' mean perceived risk from GM plants and animals as measured by the items “risky”, “risk increasing” and “high risk to future generations.” Independent samples t-tests conducted on the resultant “composite risk” mean scores revealed no significant differences between those with high levels of stress, arousal, and worry at the time of making risk judgments about GM food.

4. Discussion

Para. 6 Given that past accounts of public perceptions of GM food in the UK and Europe have been so pessimistic (Gaskell *et al.*, 2000), and media coverage largely negative (Nerlich *et al.*, 2000), the finding that 93% of our sample were willing to taste an apple that they believed to be GM is rather astonishing. It seems to suggest that worries and negativity about GM food are not as high as other studies suggest. Indeed, in the light of past research results we expected that many more participants would refuse to taste the apple. We were in fact hoping to conduct analyses to predict membership for “tasters” and “nontasters”. The fact that so few people refused to taste the apple made it impossible to conduct such tests.



Lesson 1 Reading

Para. 7 One explanation for the discrepancy between the present study and previous studies is that participants in our study were recruited “topic-blind,” meaning that those with strong interest or opinions were unable to self-select. This has been acknowledged as a limitation of past studies (Cook *et al.*, 2002).

Para. 8 We were, however, able to examine differences between individuals who said they would / would not buy GM food in the future. Surprisingly, nearly half our sample (48%) indicated that they would buy GM food when it became available. The largest differences in attitudes between purchasers and nonpurchasers were observed on the dread — not dread, and ethical — unethical, scales. It should be noted that 85.7% of nonpurchasers were happy to taste a GM apple, which suggests that the fact that someone does not intend to purchase GM food does not mean that their opposition to it is extreme. There is a spectrum of possible attitudes that people who do not intend to buy GM food can have, ranging from mild uncertainty to morbid fear to hostile aggression. The fact that someone chooses to try the GM apple suggests that they are closer to the mild end of opposition. (It is possible that we may have found more negative responses if we had used a product described as being derived from a GM animal since, as Table 1 reveals, attitudes toward GM animals tended to be slightly more negative than those toward GM plants).



Lesson 1 Reading

Para. 9 Logistic regression analysis revealed that dread (integral affect) of GM food and concerns about the future risks of GM animals used in food are key determinants of willingness to purchase GM food. This is an important finding and suggests that further investigation of integral affect or feelings toward GM food should be made. Given that our predictive model accounts for approximately half of the variance in determining willingness to purchase GM food, the effects of other important feelings, such as anger, outrage, and sadness, should perhaps be examined in the future. Of course, some of the remaining variance to be explained in predicting purchasing behavior may not relate to integral affect at all (indeed the future risk posed by GM animals was also a key predictor of willingness to purchase GM food), but may relate to other issues such as concerns about long-term effects, concerns about the environment, trust in scientists and regulatory bodies, and the degree to which GM food is perceived to be needed (Marris *et al.*, 2002).

Para. 10 Contrary to expectation, incidental affect was not found to affect ratings of the perceived riskiness of GM. On the basis of past studies it was expected that those experiencing increased levels of stress and worry would rate GM food as more risky than those experiencing low levels of incidental affect, but this was not the case.

(1392 words)



Lesson 1

References

- Cook, A. J., Kerr, G.N.& Moore, K. (2002). Attitudes and intentions towards purchasing GM food. *Journal of Economic Psychology*, 23, 557-572.
- Gaskell, G., Allum, N., Bauer, M., Durant, J., Allansdottir, A., Bonfadelli, H., et al. (2000). Biotechnology and the European public. *Nature Biotechnology*, 18, 935-938.
- Marris, C., Wynne, B., Simmons, P., Weldon, S., Carceres, J., de Marchi, B., et al. (2002). Public Perceptions of Agricultural Biotechnologies in Europe: Final Report of the PABE Research Project. Commission of European Communities.
- Nerlich, B., Clarke, D. D., & Dingwall, R. (2000). Clones and crops: The use of stock characters and word play in two debates about bioengineering. *Metaphor and Symbol*, 15, 223-239.

Extracted from *Risk Analysis*, Vol. 24, No. 5, 2004



Lesson 1 Comprehension Check

A. Skim Paras. 1-5 of Text A and match each of the following main ideas with the appropriate paragraph number.

No.	Main Ideas	Para.
1-5	No important differences can be found in appearance ratings between purchasers and nonpurchasers, but as for tastiness, nonpurchasers have lower ratings of what they believed to be the GM apple.	Para.3
	The results of analysis on attitude and affect variables showed that our model can give reasons for almost half of the variance in willingness to purchase GM food.	Para.4
	Nonpurchasers give much higher ratings for both GM plants and GM animals on most attitude scales.	Para.1
	In terms of risk judgments about GM food, no obvious differences exist between those with high and low levels of stress, arousal, and worry.	Para.5
	Due to the small number of participants refusing to taste the GM apple, it was impossible to compare these groups on other key variables.	Para.2



Lesson 1 Comprehension Check

B. Skim Paras. 6-10 and summarize the main idea for each paragraph.

Para. 6	The finding suggests that worries and negativity about GM food are not as high as other studies suggest.
Para. 7	The “topic-blind” recruitment of participants makes a difference to the results of the study.
Para. 8	Instead of an extreme opposition to GM foods, people not willing to buy GM foods have a wide range of attitudes.
Para. 9	People’s willingness to purchase GM food is greatly influenced by fear and concerns about the future risks.
Para. 10	Contrary to the result of previous studies, incidental affect did not have an impact on ratings of the perceived riskiness of GM foods.



Lesson 1 Micro-skills

- I. Understanding figures & tables
- II. Comparison and contrast (1)
- III. Use of passive voice



Lesson 1 Understanding figures & tables

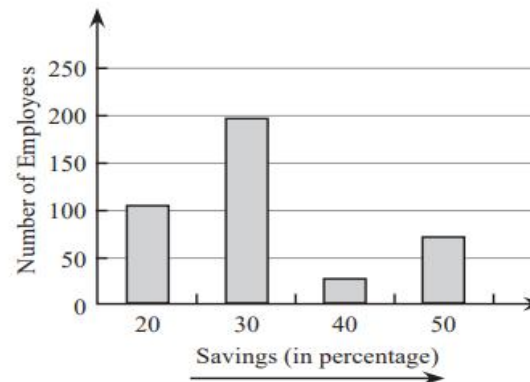
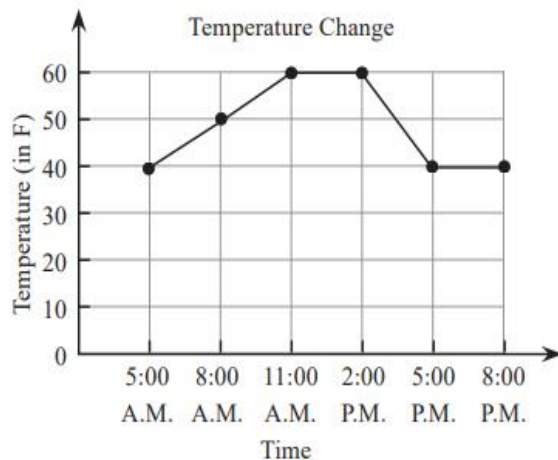
Figures and tables are very important for presenting results and data in academic writing. Well-designed, well-organized figures and tables enable readers to understand information more effectively by revealing complex relationship in a simple way. Figures (which come in the form of graphs, charts, drawings, photos, or maps, etc.) are to display trends and patterns of relationships. In a formal research paper, tables are more often used because they can illustrate raw data and explain variables.



Lesson 1 Understanding figures & tables

A. Match the diagrams with the expressions in the box.

Bar graph Line graph Pie chart Tree diagram



1. Line graph

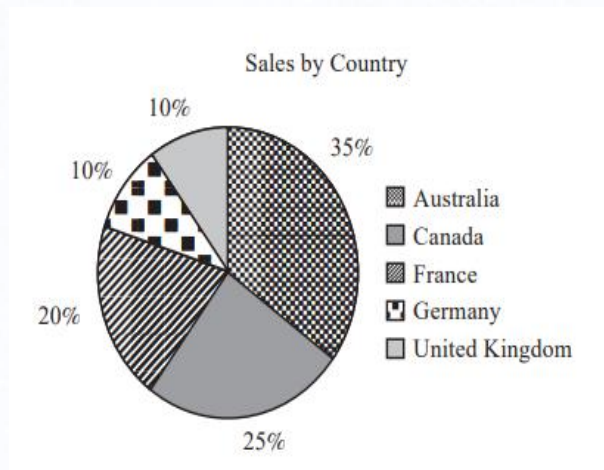
2. Bar graph



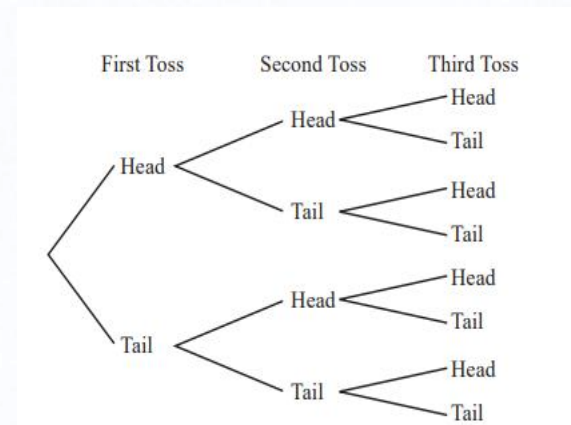
Lesson 1 Understanding figures & tables

A. Match the diagrams with the expressions in the box.

Bar graph Line graph Pie chart Tree diagram



3. Pie chart



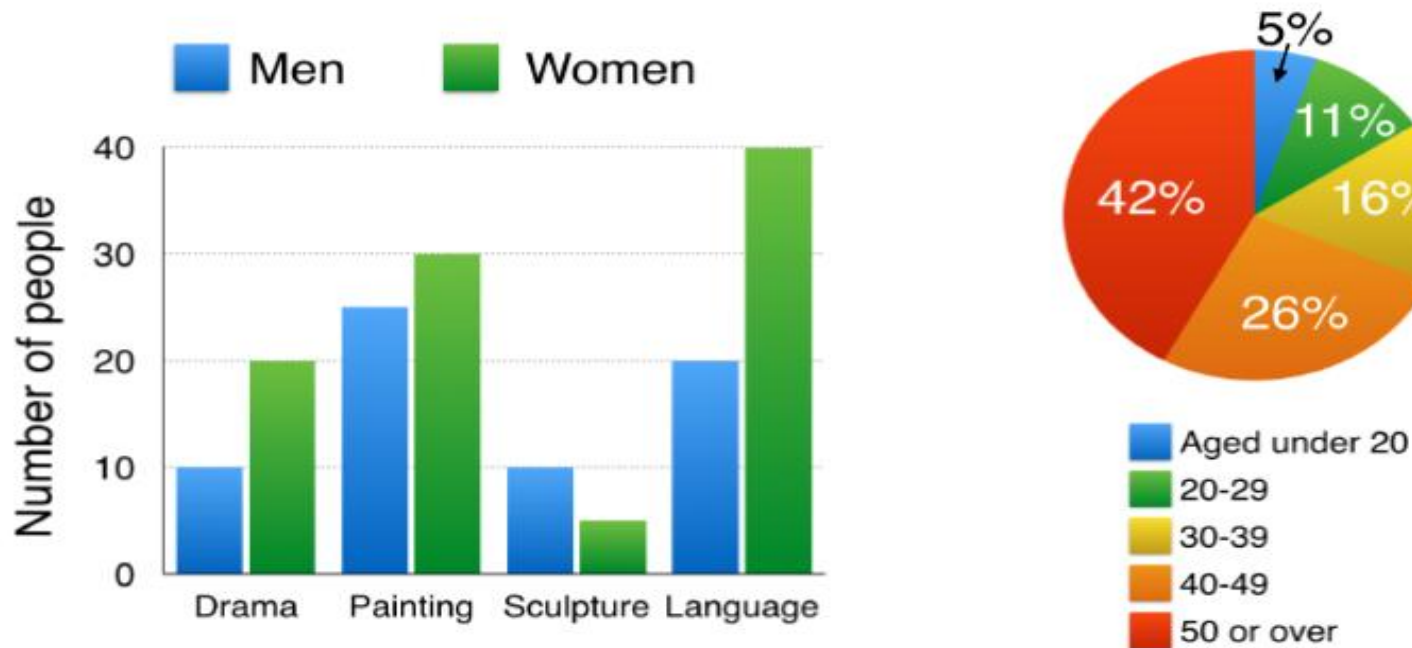
4. Tree diagram



Lesson 1 Understanding figures & tables

B. Read the bar chart and pie chart below and answer the questions.

The bar chart below shows the numbers of men and women attending various evening courses at an adult education centre in the year 2009. The pie chart gives information about the ages of these course participants.





Lesson 1 Understanding figures & tables

B. Read the bar chart and pie chart below and answer the questions.

1. What does the bar chart show? And what about the pie chart?

The bar chart compares the numbers of males and females who took four different evening classes in 2009, and the pie chart shows the age profile of these attendees.

2. Which course attracted most people?

Language course.

3. Which age group was more interested in evening courses?

50 or over.

4. Who were more interested in evening courses, men or women?

Women.

5. What was the percentage of people aged under 20 who attended evening courses?

5%.



Lesson 1 Understanding figures & tables

- C.** You should use a table when you have a list of numbers or texts. And you present that information in labeled columns. Observe the table below closely and write a report describing it.

The table below shows statistics about the top five countries for international tourism in 2012 and 2013.

Country	Number of tourists, 2012 (millions)	Number of tourists, 2013 (millions)	Tourist spending, 2012	Tourist spending, 2013
France	83.0	84.7	\$53.6 billion	\$56.1 billion
USA	66.7	69.8	\$126.2 billion	\$139.6 billion
Spain	57.5	60.7	\$56.3 billion	\$60.4 billion
China	57.7	55.7	\$50.0 billion	\$51.7 billion
Italy	46.4	47.7	\$41.2 billion	\$43.9 billion



Lesson 1

Sample

The table compares the five highest ranking countries in terms of the numbers of visits and the money spent by tourists over a period of two years.

It is clear that France was the world's most popular tourist destination in the years 2012 and 2013. However, the USA earned by far the most revenue from tourism over the same period.

In 2012, 83 million tourists visited France, and the USA was the second most visited country, with 66.7 million tourists. Spain and China each received just under 58 million visitors, while Italy was ranked fifth with 46.4 million tourists. 2013 saw a rise of between 1 and 4 million tourist visits to each country, with the exception of China, which received 2 million fewer visitors than in the previous year.

Spending by tourists visiting the USA increased from \$126.2 billion in 2012 to \$139.6 billion in 2013, and these figures were well over twice as high as those for any other country. Spain received the second highest amounts of tourist revenue, rising from \$56.3 billion to \$60.4 billion, followed by France, China and Italy. Interestingly, despite falling numbers of tourists, Chinese revenue from tourism rose by \$1.7 billion in 2013.



Lesson 1 Comparison and contrast (1)

In academic writing, we often **compare or contrast** two or more definitions, situations, events, results, etc. This strategy is more often used in the Results and Discussion sections of an academic paper since the researchers usually summarize the main findings by making generalized comparison and contrast between data. Note that it is not necessary to lay equal emphasis on every change of data. Stress should be given to those dramatic changes or those that are meaningful to further research.



Lesson 1 Comparison and contrast (1)

A. Match the examples taken from Text A (1-6) to the ways of comparison and contrast (a-c).

1, 5 a. Similarities and differences are signaled using linking expressions (e.g. similarly, while).

2, 4 b. Differences are highlighted using comparative adjectives or adverb phrases (with more / less / -er than...) or the words “more” or “less”.

3 c. Similarities and differences are not highlighted with any particular words or phrases.
In this case, clauses or sentences often have the same phrases or structure.

1. Results showed that 48% of the sample stated that they would buy GM food when it became available (purchasers), compared to 49% who would not (nonpurchasers).
2. Nonpurchasers demonstrated significantly higher (i.e. negative) ratings for both GM plants and GM animals on most attitude scales.
3. In regard to appearance, there were no significant differences between purchasers and nonpurchasers in ratings for the three kinds of apples. In regard to tastiness, purchasers and nonpurchasers are divided in ratings.
4. Nonpurchasers viewed GM food as significantly more dreaded, more risky, and of a higher risk to future generations.
5. Contrary to expectation, incidental affect was not found to affect ratings of the perceived riskiness of GM.



Lesson 1 Comparison and contrast (1)

B. Look at the table below and write a paragraph in which you select the data to make significant comparisons and give possible explanations.

Age Group	Male	Female
0-4	12.0%	4.5%
5-14	9.1%	3.0%
15-24	9.1%	8.5%
25-34	11.6%	33.6%
35-44	10.7%	13.7%
45-54	9.8%	6.7%
55-64	23.6%	19.5%

Lesson 1 Comparison and contrast (1)

Sample

The results in the table above demonstrate that in the age group of 0-4 and 5-14, the percentage of hospitalized males is nearly triple that of females. These results suggest that young boys are generally more prone to illness or accidents requiring medical care than their female counterparts. Another very sharp contrast in the data for genders is shown in the age group of 25-34. Here the hospitalization rate of females triples that of males. This difference is mainly attributed to the fact that women generally give birth during this age range. Other differences are less significant although more males than females are hospitalized when they are between 55-64. This may indicate that men suffer more than women from illness in old age.



Lesson 1 Use of passive voice

Sentences in **passive voice** are more frequently used in academic writing since they highlight the action and result rather than the person who performs the action. The use of passive voice makes the academic writing more objective and impersonal.

e.g.

1. *This magnitude of difference in ratings between the two groups **was also observed** in judgments on the risk of GM. (Para.1)*
2. *A forward (likelihood ratio) stepwise logistic regression **was conducted** on the attitude and affect measures to investigate this. (Para.4)*



Lesson 1 Use of passive voice

Rewrite the following sentences by turning active voice into passive voice.

1. We randomly assigned the participants to three groups.

The participants were randomly assigned to three groups.

2. In order to investigate the reasons for the decrease of working efficiency, we conducted a comparative study between the experiment group and the control group.

In order to investigate the reasons for the decrease of working efficiency, a comparative study was conducted between the experiment group and the control group.

3. Some researchers found progressive muscle relaxation an excellent technique to relax skeletal muscles, internal organs, and the mind.

Progressive muscle relaxation was found to be an excellent technique to relax skeletal muscles, internal organs, and the mind.

4. Researchers instructed students who opted not to complete the survey to sit quietly until all surveys were completed.

Students who opted not to complete the survey were instructed to sit quietly until all surveys were completed.

5. People have also linked acute and chronic stress to psychological and emotional problems such as anxiety, depression, irritability, frustration, anger, uncertainty, and lack of confidence. Acute and chronic stress has also been linked to psychological and emotional problems such as anxiety, depression, irritability, frustration, anger, uncertainty, and lack of confidence.



Lesson 1

Macro-skills

Writing a results section



Lesson 1 Writing a results section

A. Complete the following description about the Results section using the words in the box.

A. complement B. concise C. magnitude D. relevant
E. heading F. categories G. consecutively H. past tense

Functions and contents: The results of your survey, investigation and experiment are presented in the Results section. It is the core of the research in which you will summarize the main findings from the data and make generalized comparisons.

Steps to writing an effective Results section:

1. Determine which results to present by deciding which are D to the questions raised in the Introduction.
2. Group the data into F in order of importance. Grouping permits a smooth and logical reporting of the data.
3. Highlight significant findings and guide the readers to the relevant data in the text. The text should A figures and tables, instead of repeating the same information.



Lesson 1 Writing a results section

A. complement B. concise C. magnitude D. relevant
E. heading F. categories G. consecutively H. past tense

4. Provide a clear description of the C of a response or difference. If appropriate, use percentage of change rather than exact data.
5. Use the H when you refer to your results since the study is finished.
6. Number figures and tables G in the same sequence they are first mentioned in the text.
7. Provide a E for each figure and table, which could stand on its own, separate from the text.
8. Write in a B, accurate and clear manner.



Lesson 1 Writing a results section

B. The following extract is taken from the results section of an academic paper, which describes information in Tables. Put the sentences (1-5) in the correct order.

1. Respondent data were grouped as “very unlikely” (scores of 2 or less) or “very likely” (scores of 4 or higher) for analysis.
2. In addition, a majority (73.6%) of the respondents expect PA to be fun.
3. Questions concerning exercise outcome expectations consist of Likert-type scales ranging from 1 (very unlikely) to 5 (very likely).
4. Responses to questions concerning positive and negative outcome expectations for PA revealed that, overall, students expect PA to promote and protect their health, help them look and feel better and manage stress.
5. The highest scores for negative outcome expectations associated with PA included soreness and time to get ready and clean up after PA .



Lesson 1 Writing a results section

Positive outcome expectations for physical activity^a

Promote health	Very unlikely	Very likely
Benefit the health of my heart	4.8%	79.8%
Improve my overall health	0%	89.4%
Give me energy	8.6%	57.1%
Help me live long	3.8%	73.6%
Enhance appearance		
Help me in controlling weight and improving appearance	7.8%	75.5%
Help me get or keep in shape	2.9%	82.9%
Stress management / mental health		
Help me feel good about myself	4.9%	75.7%
Help me cope with stress and tension	15.7%	56.9%
Help me avoid feeling guilty	29.5%	40.0%
Result in having fun	3.8%	73.6%

^aHow likely would the following occur if you were to be physically active on a regular basis during the next 3 months?



Lesson 1

Writing a results section

Negative outcome expectations for physical activity^a

	Very unlikely	Very likely
Lead to my getting injured	67.3%	8.5%
Cause me pain and muscle soreness	23.3%	34.0%
Result in my overexerting myself	40.4%	17.3%
Make me tired	27.4%	29.2%
Result in my being embarrassed	69.2%	12.5%
Interfer with my other responsibilities	48.6%	15.1%
Involve a lot of getting ready and cleaning up	23.3%	33.0%

^aHow likely would the following occur if you were to be physically active on a regular basis during the next 3 months?



Lesson 1 Writing a results section

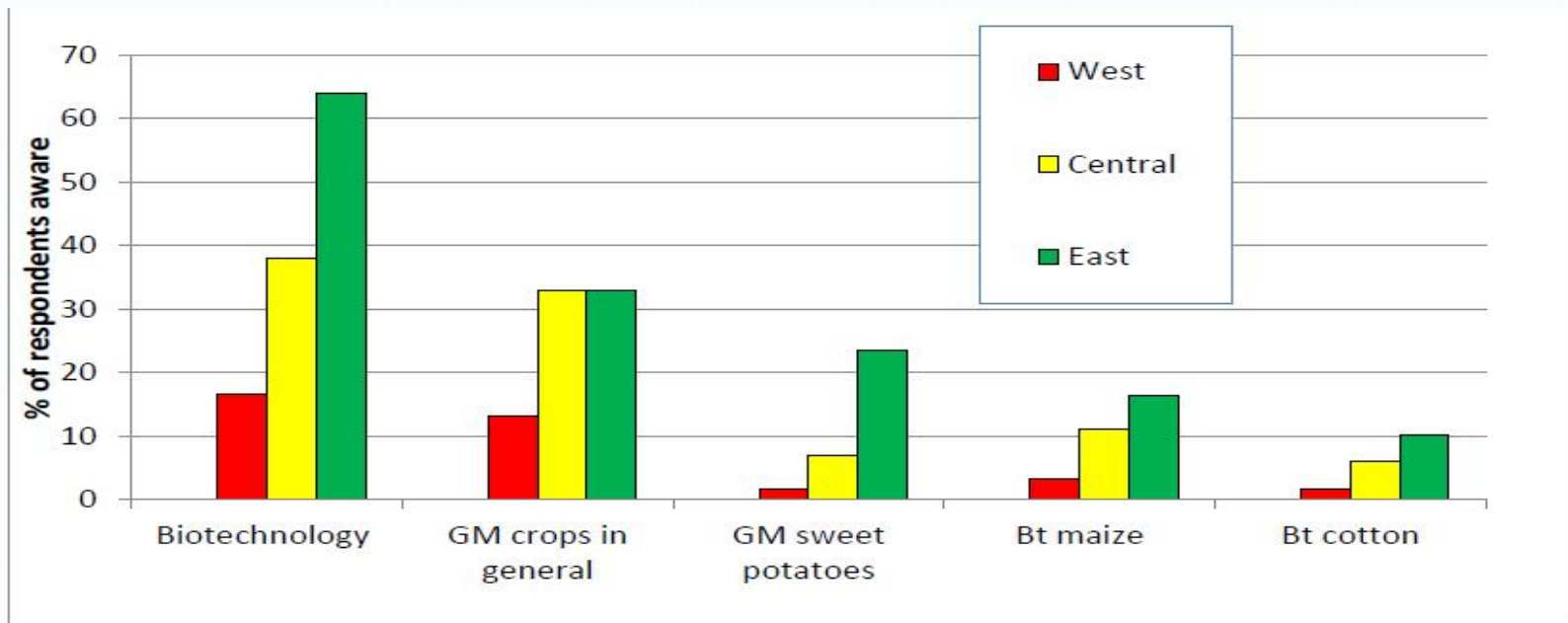
3. Questions concerning exercise outcome expectations consist of Likert-type scales ranging from 1 (very unlikely) to 5 (very likely).
1. Respondent data were grouped as “very unlikely” (scores of 2 or less) or “very likely” (scores of 4 or higher) for analysis.
4. Responses to questions concerning positive and negative outcome expectations for PA revealed that, overall, students expect PA to promote and protect their health, help them look and feel better and manage stress (Table 3).
2. In addition, a majority (73.6%) of the respondents expect PA to be fun.
5. The highest scores for negative outcome expectations associated with PA included soreness and time to get ready and clean up after PA (Table 4).



Lesson 1 Writing a results section

C. Look at the bar chart below which shows the results of a survey about awareness of biotechnology and GM technology in Kenya. Fill in the blanks using the words from the box.

GM sweet potatoes Central more
biotechnology Western a third



Awareness of rural consumers of biotechnology



Lesson 1 Writing a results section

As can be seen in the Figure , less than half the respondents in the different surveys were aware of biotechnology, although this differed substantially between groups. In the East, more than 60% were aware of 1. biotechnology but fewer respondents in 2. Central (38%) and 3. Western (16%) zones were aware. The respondents were generally 4. more aware of biotechnology than about GM crops. The zone closer to urban centers were substantially more aware of GM crops (about 5. a third in Central and in the East), than those further away (13% in the West). Within the GM crop category, respondents were more aware of 6. GM sweet potatoes, followed by Bt maize and Bt cotton, likely reflecting their history in Kenya.



Lesson 1 Writing a results section

D. Useful expressions in the results section

1. Learn about some sentence patterns showing the results/findings.

As shown/can be seen in Figure 1...

Table 1 shows/illustrates/suggests that...

The results of this analysis may be seen in Table 2.

The percentage/ number of ...has almost doubled as compared with...

There was a(n) X percent increase/decline in the number/rate...

There were large differences in...

No significant differences were observed / found between...

The results indicate/suggest that...



Lesson 1 Writing a results section

2. Underline the expressions used to describe results in the results section of Text A.

- 1) Results showed that 48% of the sample stated that they would buy GM food when it became available (purchasers), compared to 49% who would not (nonpurchasers), with three nonresponses. (Para.1)
- 2) Table 1 illustrates the magnitude of differences between the two groups. (Para.1)
- 3) There were large differences in ratings of whether genetic modification of plants and animals for food is dreaded and viewed as unethical, with nonpurchasers' ratings at least 22% higher on these scales for both plants and animals. (Para.1)
- 4) No differences were observed between purchasers and nonpurchasers' ratings in regard to the perceived dullness — vividness of the topic of GM food, and both groups felt that GM animals and plants affected them personally. (Para.1)
- 5) A mixed-design ANOVA revealed there were no significant differences between purchasers and nonpurchasers in regard to appearance ratings for the “organic”, “traditional” and “GM” apples. (Para.3)
- 6) The results of this analysis may be seen in Table 2. (Para.4)



Lesson 1

- **Lead-in**
- **Reading**
- **Critical Thinking**
- **Micro-skills**
- **Macro-skills**



Lesson 1

Lead-in

- A. Regular PA (Physical Activity) is usually viewed as an effective way to maintain health and keep in shape. What PA courses (Physical Education lessons) have you enrolled in? How long does each activity last? Complete the following table and share it with your partner.**

List of PA courses	Duration for each course per week (minutes)



Lesson 1

Lead-in

B. Before you start reading the results section of the academic article *Activity Levels of College Students Enrolled in Physical Activity Courses*, discuss the following questions with your partner.

1. Do you think the time you spend on PA is sufficient to maintain health?
2. What is your primary reason for taking PA courses? You may choose one from the following:
 - a. gaining knowledge of the sport
 - b. developing skills in various sports
 - c. fulfilling a requirement
 - d. increasing physical activity
 - e. controlling weight
 - f. other reasons
3. What advice would you give for a stronger college/university PA program?



Lesson 1

Reading

Text B

Activity Levels of College Students Enrolled in Physical Activity Courses (Extract 2)

Results

PA and sedentary time across all time points

Para. 1 Figure 1 shows the results for time spent in sedentary behavior, light PA, and moderate- to-vigorous PA (MVPA) for each class across all time points. A total of 54 class periods were observed. Classes in this study were either offered for 90-minute periods (archery, flying disc sports, Tai Chi, yoga) for a total possible measurement time of 540 minutes or 50-minute periods (jogging, karate, spinning, swing dance, weight training) with a total possible measurement time of 300 minutes.



Lesson 1

Reading

Within the 90-minute classes, Tai Chi averaged the most sedentary time (68%, 61.4 min/class, SD=9.74 min), archery averaged the most time in light PA (45%, 40.0 min/class, SD=3.48 min), and flying disc sports averaged the most time in MVPA (22%, 20.0 min/class, SD=2.15 min). The 90-minute class period that yielded the highest MVPA at a single time point was flying disc sports with 47 minutes (52%) spent in MVPA. Within the 50-minute classes, swing dance averaged the most sedentary time (60%, 30.1 min/class, SD=11.25 min), spinning averaged the most time in light PA (54%, 27.0 min/class, SD=4.29 min), and jogging averaged the most time in MVPA (39%, 19.5 min/class, SD=5.80 min). The 50-minute class that yielded the highest MVPA at a single point was weight training with 37.5 minutes (75%) spent in MVPA.

Lesson 1

Reading

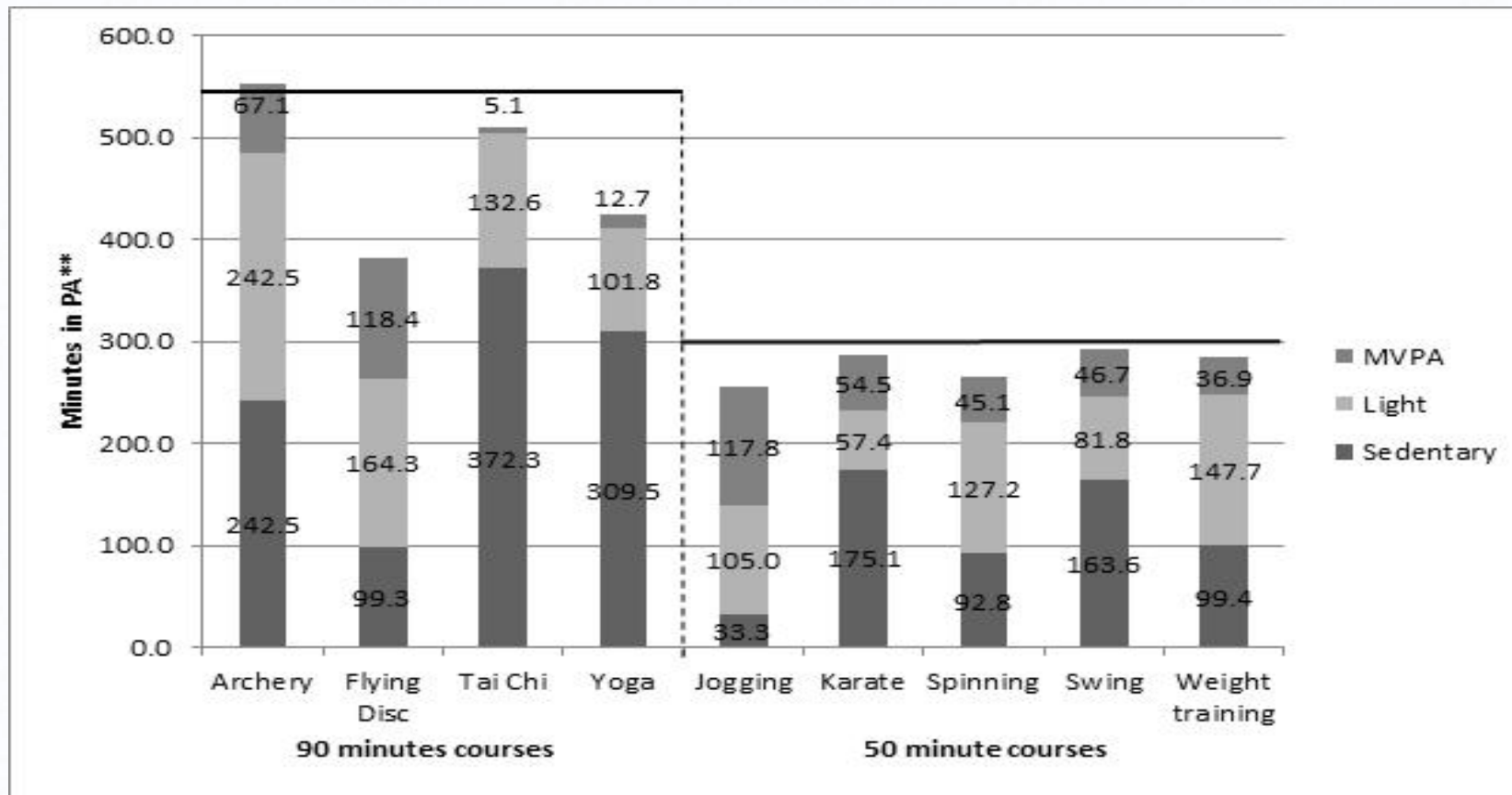


Fig. 1 Total time in PA by class across all six time points

Notes:*Solid line indicates maximum total time scheduled.

**Validated adult hip-based cut points using Freedson et al. (1998) in 10s EPOCH; sedentary (<16.5), light (16.6-325.1), moderate (325.2-954), and vigorous intensity (>954.1) PA.



Lesson 1

Reading

PA and sedentary time at each time point

Para. 2 Table 1 presents the results for time spent in sedentary, light PA and MVPA for each class by time point. We present here the time point which represents the greatest amount of MVPA. For the 90-minute classes, at time point three, flying disc sports had the most time (45%, 41 minutes) spent in MVPA, while Tai Chi had the least at time point one with zero minutes of MVPA. At time point three, students in archery class spent 42 minutes (46%) in light PA, while at time point three Tai Chi yielded only 8 minutes (9%) of light PA which is the lowest light PA score for the 90 minute courses. Students in Tai Chi also spent the most time (85%, 77 minutes) being sedentary at time point three, while the least amount of time spent being sedentary was in flying disc sports at time point five (11%, 10 minutes). For the 50-minute classes, jogging had the most time (50%, 25 minutes) spent in MVPA during time points two and six, while swing had the least at time point three with just two minutes (4%) of MVPA. At time point three, students in spinning class spent 33 minutes (66%) in light PA, while at time point five weight training yielded only one minute (2%) of light PA which is the lowest light PA score for the 50 minute courses. Students in weight training also spent the most time (88%, 44 minutes) being sedentary at time point five, while the least amount of time spent being sedentary was during jogging at time points two and six (2%, 1 minute).



Lesson 1

Reading

Reasons for enrolling in PA classes

Para. 3 The following results are presented as totals across all classes. Participants selected the following response options as reasons why they enrolled in the PA classes: (a) fulfill a requirement (n=13), (b) gain knowledge of the sport/skill (n=26), (c) increase PA (n=25), (d) weight control (n=4) and (e) other (n=11). The majority of students (76%, n=50) reported choosing to take the course as an elective, and 44% (n=29) of students reported taking the course as either a means of weight control or to increase their PA levels.

Discussion

Para. 4 College PA programs are a viable means for providing young adults with increased PA opportunities and education that can help to promote the knowledge, competence and confidence needed for sustained, out-of-class PA participation and a healthy lifestyle (De Meester, Aelterman, Cardon, De Bourdeaudhuij, & Haerens, 2014). Currently, there are no guidelines or recommendations specific to the number of minutes that students should spend in PA, at any level of intensity, during college PA class periods. The recommendation for K-12 physical education is for students to spend 50% of the class period in MVPA (Wiecha, Hall, & Barnes, 2014). In this study, students were engaged in MVPA for at least 50% of the class period during



Lesson 1

Reading

only three of the observed class periods (once during a swing class and twice during a jogging class. Weaver, Webster, and Beets (2013) recommend using the LET US Play principles (LET US is an acronym for Lines, Elimination, Team size, Uninvolved teachers and students, and Space, equipment, and rules) to maximize PA opportunities during physical education lessons. In a recent study, the application of these principles during physical education lessons taught to children increased the amount of time children spent in MVPA (Brazendale, Chandler, *et al.*, 2015; Weaver, Webster, & Beets, 2013). Additionally, another intervention utilizing the LET US Play principles had the elementary physical education teachers incorporate the principles in their planning and teaching with the intention of eliciting an increased accumulation of MVPA (Weaver, Webster, Egan, Campos, & Michael, in press). Introducing college PA instructors to these principles (via orientations, trainings, booster sessions, and monitoring) may promote more physically active classes.



Lesson 1

Para. 5 Only one (flying disc sports) of the possible 54 class periods accumulated enough PA to meet the daily goals of a minimum of 30 minutes of MVPA a day (Haskell *et al.*, 2007). As previously stated, the primary goal of the PA program described in this study is to help students become physically educated so that they may pursue a lifetime of health and PA. The PA patterns which are established during a student's college years are very likely to be maintained into adulthood and have a long-term impact on adult PA habits (Fish, 1996; Sparling & Snow, 2002). The American College Health Association-National College Health Assessment II (2013) identifies physical inactivity as a priority health risk behavior for college populations.



Lesson 1

Reading

College students are adults with multiple responsibilities and live a lifestyle that mimics adulthood and as a result they are likely to maintain PA patterns that are established during their time in college (Calfas, Sallis, Lovato, & Campbell, 1994). The role of the higher education PA instructor includes the challenge of encouraging college students to participate in recommended amounts of PA which may help students establish habitual participation in PA (Wallace, Buckworth, Kirby, & Sherman, 2000). However, higher education institutes currently do not encourage students to become physically active adults (Keating *et al.*, 2005). A recent study of children attending afterschool programs suggested that providing more than 60 minutes per day for PA opportunities did not result in further accumulation of MVPA and that an optimal zone (i.e. PA opportunity time) of 60-75 minutes for the accumulation of MVPA exists for both sexes (Brazendale, Beets, *et al.*, 2015). The solution to the problem of increasing the amount of time students spend in MVPA may be addressing college PA instructors class structure and delivery techniques (e.g. using LET US Play principles) in an effort to maximize existing class time for optimal MVPA accumulation.



Lesson 1

Reading

Para. 6 In addition to describing the amount of PA students accumulate during college PA classes, this study identified students' reasons for enrolling in college PA courses (Brazendale, Beets, *et al.*, 2015). Results indicated students primarily enroll in these courses to increase their knowledge of the activity (38%) or to increase PA (36%). These results are similar to previous studies that have reported on college PA courses (Broer & Holland, 1954; Leenders *et al.*, 2003; Savage, 1998). In contrast, large portions of the National College Health Risk Survey (American College Health Association, 2010) and others reported that college students exercise to lose weight (Kilpatrick, Hebert, & Bartholomew, 2005; Lowry *et al.*, 2000) Since the classes included in this study ranged from skill courses such as archery and flying disc sports to more traditional fitness courses such as spinning and yoga there may exist a disparity in the reasons for taking a particular type of course.

(1314 words)



Lesson 1

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Lesson 1 Critical Thinking

Read the text again and answer the following questions.

1. How may the LET US PLAY principles help promote students' PA opportunities?

They may increase the amount of time students spent in MVPA.

2. Why is it important to establish an ideal PA pattern during a student's college time?

Because students are likely to maintain this pattern into adulthood.

3. What are the major reasons for students to enroll in PA courses according to the studies?

They aim to increase knowledge of activity, to increase PA and to lose weight.



Lesson 1

Micro-skills

I. Comparison and contrast (2)

II. Hedging



Lesson 1

Comparison and contrast (2)

One of the elements commonly found in the Discussion section of a research paper is to compare your results with those of previous researchers. Your results may either support or differ from those of previous studies.

e.g.

“These results are similar to previous studies that have reported on college PA courses ” (Para. 6, Text B).

Often, possible reasons may be offered when the results are different.

e.g.

“One explanation for the discrepancy between the present study and previous studies is that participants in our study were recruited ‘topic-blind’ ” (Para. 7, Text A).



Lesson 1

Comparison and contrast (2)

Decide whether the researcher's results agree (write A) or disagree (write D) with those of previous researchers. Underline the phrases that indicate agreement or disagreement.

- D 1. These results contrast with those of Ackerman and Gross (2009), who found that students with less free time to study had a significantly higher GPA than those with more.
- A 2. Our findings accord with earlier research indicating, for example, that students' study time is increasing.
- A 3. The present findings confirm previous reports (Schmid, 2002) of no relationship between time spent studying outside of class and student learning.
- A 4. These results are consistent with the previous studies on the time freshmen spent per week outside of class on academic work by Young (2016).
- D 5. Bennet (2003) found no relationship between work and academic performance. Conversely, in Goddard's (2003) study of 1,500 students in seven universities, work was found to interfere with the acquisition of academic knowledge and skills.
- D 6. The results of this qualitative study differ from those reported by Murphy (2006).



Lesson 1

Hedging

In academic language it is very common to avoid making statements that are too direct or too confident. This is because in academic world some of the ideas discussed are not facts that we are sure about. We can avoid directness by using special words and expressions which indicate that something may be true, is partially true, is true in some cases. This is known as **hedging**. Hedging language is often included in claims and other ideas in the discussion section of a research paper. Hedging can reduce the strength of a claim and make it more precise. It can also help maintain an attitude of objectivity, which is often associated with the writer's credibility in a research paper.



Lesson 1 Hedging

There are a number of types of hedge, including:

a. Modal verbs	e.g. can, may, might, could
b. Adverbs and adverbial phrases	e.g. typically, probable, possibility, in most cases, to some extent, likely to
c. Verbs	e.g. seem, appear, assume
d. Impersonal structures with <i>It</i> and <i>There</i>	e.g. It is believed that ..., It is possible ..., There seems to ..., There is a possibility that ...



Lesson 1 Hedging

A. Underline the hedges in the extracts from Text A & Text B and identify which type of hedge (a-d) they belong to.

- d 1. It seems to suggest that worries and negativity about GM food are not as high as other studies suggest. (Para. 6, Text A)
- d 2. It is possible that we may have found more negative responses if we had used a product described as being derived from a GM animal. (Para. 8, Text A)
- b 3. Given that our predictive model accounts for approximately half of the variance in determining willingness to purchase GM food, the effects of other important feelings, such as anger, outrage, and sadness, should perhaps be examined in the future. (Para. 9, Text A)
- a 4. Introducing college PA instructors to these principles (via orientations, trainings, booster sessions, and monitoring) may promote more physically active classes. (Para. 4, Text B)
- b 5. The PA patterns which are established during a student's college years are very likely to be maintained into adulthood and have a long-term impact on adult PA habits (Fish, 1996; Sparling & Snow, 2002). (Para. 5, Text B)



Lesson 1 Hedging

B. Add a hedge from the box to the following statements where appropriate.

mainly	often	It is widely accepted that
generally speaking	relatively	are likely to

1. Cities in the Northern Hemisphere become hotter over the next century.
Cities in the Northern Hemisphere are likely to become hotter over the next century.
2. Air pollution has got worse in developing countries because of economic growth.
Air pollution has got worse in developing countries mainly because of economic growth.
3. Maps that once had to be hand-drawn can now be created on a computer and printed in a short amount of time.
Maps that once had to be hand-drawn can now be created on a computer and printed in a relatively short amount of time.



Lesson 1 Hedging

B. Add a hedge from the table to the following statements where appropriate.

mainly

often

It is widely accepted that

generally speaking

relatively

are likely to

4. What we think of as a single innovation is the result of a lengthy process.

What we think of as a single innovation is often the result of a lengthy process.

5. Computers and information technology provide many benefits.

Generally speaking, computers and information technology provide many benefits.

6. Language affects one's perception of the world.

It is widely accepted that language affects one's perception of the world.



Lesson 1

Macro-skills

Writing a discussion section



Lesson 1 Writing a discussion section

The discussion section is generally regarded as the most difficult part in writing an academic paper. This section aims to comment and interpret the findings of your research and lead to the conclusion. It mainly consists of the following components:

1. restating the major findings: discussing the relationships and generalizations shown by the results;
2. explaining your findings: giving the judgment, evaluation and correlation shown by major findings;
3. comparing your results with those of previous studies: pointing out any exceptions or unsettled points;
4. identifying the significance: stating the theoretical or practical implications of your research;
5. pointing out potential limitations or the directions of future research.



Lesson 1 Writing a discussion section

A. Read the following discussion section of an academic paper and match them to the functions listed in the box.

- A. Restating major findings
- B. Offering explanations
- C. Comparing the present study with previous studies
- D. Identifying the significance or implications
- E. Discussing limitations or future research



Lesson 1 Writing a discussion section

A. Read the following discussion section of an academic paper and match them to the functions listed in the box.

1. Function: A

This study examined the cross-sectional associations between PA, weight status, and PWB in a large, representative sample of Korean adolescents. Though the associations between weight status and PWB varied by weight groups and tended to be small, PA was consistently and favorably associated with happiness and low/ no stress in all weight groups after adjusting for covariates. Furthermore, PA was favorably associated with happiness and low/ no stress in a dose dependent manner after further adjusting for weight status.

2. Function: C

Since most existing cross-sectional and longitudinal studies focused on examining negative psychological states, such as depressed symptoms (Goldfield *et al.*, 2011; Johnson *et al.*, 2008; Toseeb *et al.*, 2014) or depression (Wiles, Hasse, Lawlor, Ness, & Lewis, 2012), our findings suggest the importance of PA for positive aspects of PWB.



Lesson 1 Writing a discussion section

3. Function: D

As expected, overweight adolescents, compared to their normal weight counterparts, were less likely to be happy and more likely to feel stressed. However, participating in any type of PA at least once per week was associated with higher odds of reporting happiness and less/no stress in all three weight groups (i.e. underweight, normal weight, and overweight). These results suggest that overweight adolescents may be at risk of feeling unhappy and stressed compared to their normal weight peers, but participating in PA may lower the risk of having negative PWB.

4. Function: A

Our findings also showed that underweight adolescents are more likely to be happier and less/not stressed than their normal weight counterparts after adjusting for demographic and developmental factors.



Lesson 1 Writing a discussion section

5. Function: B

These findings may be explained by desired body ideal among teenagers in the contemporary Korean society (Lim & Wang, 2013). Similar to adolescents in Western, developed countries, thin and lean body type is preferred among Korean adolescents (Kim, Yun, & Kim, 2015). Such body ideals may also explain our results showing that underweight adolescents are more likely to be happier and less/not stressed compared to their normal weight peers. Similarly, overweight adolescents are more susceptible to experiencing negative psychological issues including low self-esteem and body dissatisfaction (Kim *et al.*, 2015).

6. Function: E

These observations can be confirmed in future studies using more rigorous study design (i.e. longitudinal, experimental).



Lesson 1 Writing a discussion section

7. Function: D

Strengths of this study include the large sample size in which provides representative estimates of the population. In addition, this study provided additional evidence in the understudied population group. Furthermore, by including happiness and stress as outcomes, this study expands our understanding of the associations between PA and PWB among adolescents.

8. Function: E

Nevertheless, this study is not without limitations. First, a self-reported PA measure was employed which may not be as strong as direct or objective measures. Furthermore, though we observed the clear association between PA and PWB after adjusting for weight status, we were not able to infer causality. Last, though we found associations between PA and PWB, the effects were small (Cohen's $d < 0.2$). Longitudinal and experimental observations are warranted to build on our findings.



Lesson 1 Writing a discussion section

B. Read the discussion section of Text B and complete the table below by identifying the paragraphs in which the components of discussion are included.

Contents	Text B
Major findings	Para. <u>4,6</u>
Explanations	Para. <u>4,5,6</u>
Comparison with other studies	Para. <u>4,6</u>
Significance or implications	Para. <u>5</u>
Limitation or future research	Para. <u>5</u>



Thanks for your attention